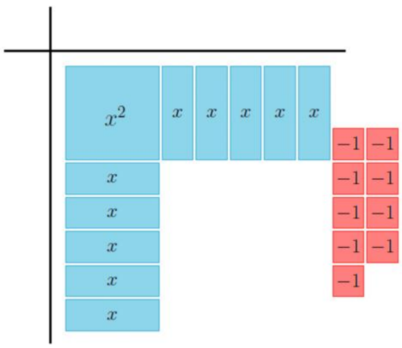
Algebra 1: Unit 8, Lesson 5

1. Three examples of equivalent quadratic equations, where $a = 1$, are shown.

	Example 1	Example 2	Example 3
Standard Form	$y = x^2 - 22x + 124$	$y = x^2 + 20x + 92$	$y = x^2 - 18x + 74$
Vertex Form	$y = (x - 11)^2 + 3$	$y = (x + 10)^2 - 8$	$y = (x - 9)^2 - 7$

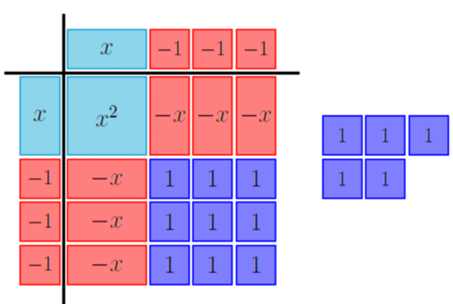
What is the relationship between the blue number in standard form (b) and the blue number in vertex form (h)?

2. The algebra tiles shown on the product mat represent the expression $x^2 + 10x - 9$.



What algebra tiles are needed to complete the square?

3. The algebra tiles shown on the product mat represent the trinomial $x^2 - 6x + 14$.



Determine if $(x + 3)^2 + 5$ is equivalent to $x^2 - 6x + 14$.